

# AI results that arrive too late: five things to measure

What a 4.5-year, 20-center study of multi-vendor AI in outpatient radiology means for heads of radiology, CMIOs and network managers

Sergey Morozov, MD, PhD, MPH

## WHAT THE STUDY FOUND

Over 4.5 years, a 20-center outpatient radiology network ran ten AI applications from seven vendors on about 389,000 examinations, and we measured when each AI result reached the radiologist. The median time from end of scan to AI result was 2.06 minutes, 72% of it spent routing data rather than running the algorithm, and in 7.2% of examinations the result arrived after the report was already finalized ("Too Late"). Where AI was integrated into the reading workflow, reports were finalized faster (radiologist-adjusted median turnaround time -26% for trauma radiography and -18% for knee MRI), yet radiologists rated their productivity gain at 2.94 out of 5, below the neutral midpoint.

**7.2%**

of AI results arrived after the report was finalized

**72%**

of total latency was data routing, not the algorithm

**-26%**

adjusted report turnaround time, trauma radiography

**2.94 / 5**

perceived productivity gain, below neutral

## FIVE THINGS TO MEASURE

### 1 Measure your Too Late rate

#### HOW TO MEASURE

For every AI-processed examination, compare two HL7 timestamps: when the AI result became available and when the report was finalized. The Too Late rate is the share of examinations where finalization came first.

#### WHY

In the study it was 7.2% overall, from 3.0% for knee MRI to 13.2% for chest CT. CT and MRI results arrived during or after finalization more often than radiography results (29.6% vs 13.6%).

### 2 Set a time-to-display target before you buy or deploy

#### HOW TO MEASURE

Agree, per modality, the maximum minutes from end of scan to AI result on the reading screen. Write it into the contract and the acceptance test.

#### WHY

This is the "Plan" step of the governance framework proposed in the paper: define time-to-display targets before implementation. Median total latency ranged from 1.81 minutes for trauma X-ray to 13.10 minutes for chest CT.

### 3 Look at routing, not only the algorithm, and prefer automated push

#### HOW TO MEASURE

Split total latency into routing (fetching, upload, download) and inference. Ask your IT team and each supplier for both numbers. Send studies to AI automatically at acquisition.

#### WHY

72% of total latency was data routing. Fetching the study from PACS was the largest single step in five of nine solutions. The paper's "Do" step is to implement automated push architectures.

## 4 Review latency and drift every quarter

### HOW TO MEASURE

Plot the quarterly median latency of each AI solution and review it with radiologist feedback. Decide what to fix, upgrade or switch off.

### WHY

Five of nine solutions were steady. The others showed progressive latency drift, post-deployment convergence, and a temporal infrastructure incident with full recovery. The paper's "Check" step is to monitor latency, algorithm drift and NPS quarterly.

## 5 Survey your radiologists twice a year

### HOW TO MEASURE

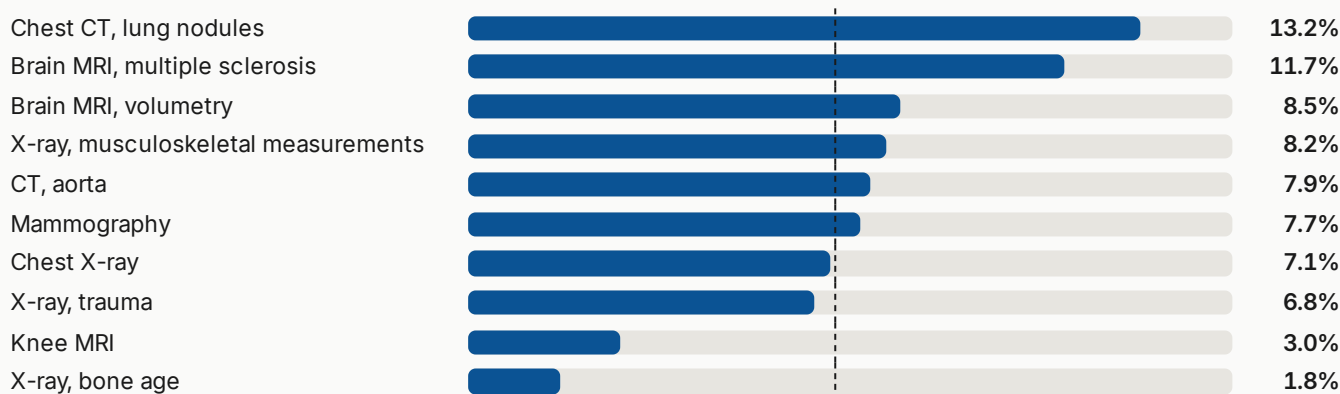
Use a short questionnaire: usage frequency, trust, quality and productivity on a 1 to 5 scale, and a 0 to 10 recommendation score for each tool.

### WHY

Perceived productivity rose from 2.57 to 2.94 but stayed below neutral, although adjusted turnaround time fell by 26% and 18%. In exploratory analysis the recommendation score for chest CT fell from +38 to -3.

**Regulation.** The paper notes that all AI applications in the study fall under the EU AI Act (Regulation 2024/1689) high-risk classification, which mandates post-market monitoring aligned with these principles.

### TOO LATE RATE BY AI APPLICATION



Share of examinations in which the AI result arrived after the report was finalized. All applications: 7.2% (1,584 of 21,872 examinations with a linked report timestamp). Dashed line: 7.2% overall.

### ABOUT THE EVIDENCE

- Observational and retrospective, in one privately managed outpatient network in French-speaking Switzerland. Results may differ in academic, inpatient or non-European settings, and with other integration architectures.
- **Associations, not causation.** The comparison group without AI (2,429 of 20,909 examinations) was defined by consent refusal and technical failure, not by random assignment.
- **Results are ceilings.** The Too Late rate shows potential loss, not measured loss: results were displayed automatically and the study did not record whether radiologists looked at them.
- Survey analyses were exploratory, with no correction for multiple comparisons. The network has a royalty relationship with one AI supplier (knee MRI), disclosed in the paper.

Morozov S, Heracleous N, Novarina O, Korka D, Dufour B, Thouly C, Rizk B. AI Latency, Report Turnaround Time, and Adoption in a Multi-Vendor AI Ecosystem: A Multi-Site Observational Study. J Am Coll Radiol. 2026. doi:10.1016/j.jacr.2026.09.026  
Research page: [aimonitoring.drsergeymorozov.com](https://aimonitoring.drsergeymorozov.com)

Sergey Morozov, MD, PhD, MPH

Questions about the study: [dr.morozov.sergey@gmail.com](mailto:dr.morozov.sergey@gmail.com)